



- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of E and C . Include appropriate units.

Data: $A = (2.2 \pm 0.2) \text{ mF}$

$E = \dots\dots\dots$

$C = \dots\dots\dots$

[2]

- (ii) Determine the percentage uncertainty in your value of C .

percentage uncertainty =% [1]

- (e) The experiment is repeated with 10 capacitors, each of capacitance C , connected in parallel between P and Q. Determine the maximum potential difference V between P and Q.

$V = \dots\dots\dots V$ [1]

[Total: 15]

